

Making a soil core extractor

Items list

Item name	Description	Link
PVC or ABS piping	Size and width adjusts to fit inside of soil corer	https://www.homedepot.ca/product/ipex-cell-core-3-inch-x-12-feet-abs-dwv-pipe/1000120780 Or https://www.homedepot.ca/product/ipex-cell-core-1-1-2-inch-x-3-feet-abs-dwv-pipe/1000423248
Flush Bushing Spg x Hub	Pipe fits into bushing to connect to Flang	https://www.homedepot.ca/product/lesso-3-x-1-1-2-in-abs-flush-bushing-spg-x-hub/1000116124
Closet Flange	Flange acts as stand to hold piping fit with bushing	https://www.homedepot.ca/product/lesso-abs-4x3-adjustable-closet-flange-spg/1000116239
Cap hub	Fits on top of pipe to create flat top	https://www.homedepot.ca/product/lesso-1-1-2-in-abs-cap-hub/1000176863 And https://www.homedepot.ca/product/lesso-test-cap-11-2-polyethylene-/1000140426
Wooden base	Any piece of flat wood to attach flag to to extend base	
Screws and bolts		https://www.homedepot.ca/product/paulin-1-4-inch-x-2-inch-pro-pack-black-carriage-bolts-5-sets-/1001766168

(Optional) - Water proof plug cap	See below	https://www.humboldtmg.com/shelby-tube-expansion-packers.html
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Instructions:



Step 1 - Pipe fitting fits into Bushing

Step 2 - Pipe+ Bushing fits into Flange. Flange is on the ground, Bushing fits inside Flang, Pipe fits inside bushing

Step 3 - Cap sits on top of pipe - adjust cap size to ensure it fits inside soil corer

Step 4 - Securing Flange to Wooden base using bolts

- Place flange on wooden base
- Using a marker or pencil, mark points for bolts through the bolt gaps in the flange
- Drill 1/4" holes to match the size of the bolts in the wood
- Place Flang on wood with holes in flange overlapping with holes in wood
- Puch bolts from the button side of the wooden base, through the flange, secure with a nut

(Optional) - Water proofing

Why - creates a tight seal between the corer and the extractor for water logged sediments



Shelby tube expansion packer



Expansion plug



Aquatic instruments custom plug



Bolt and Wing nut



Cap

Instructions

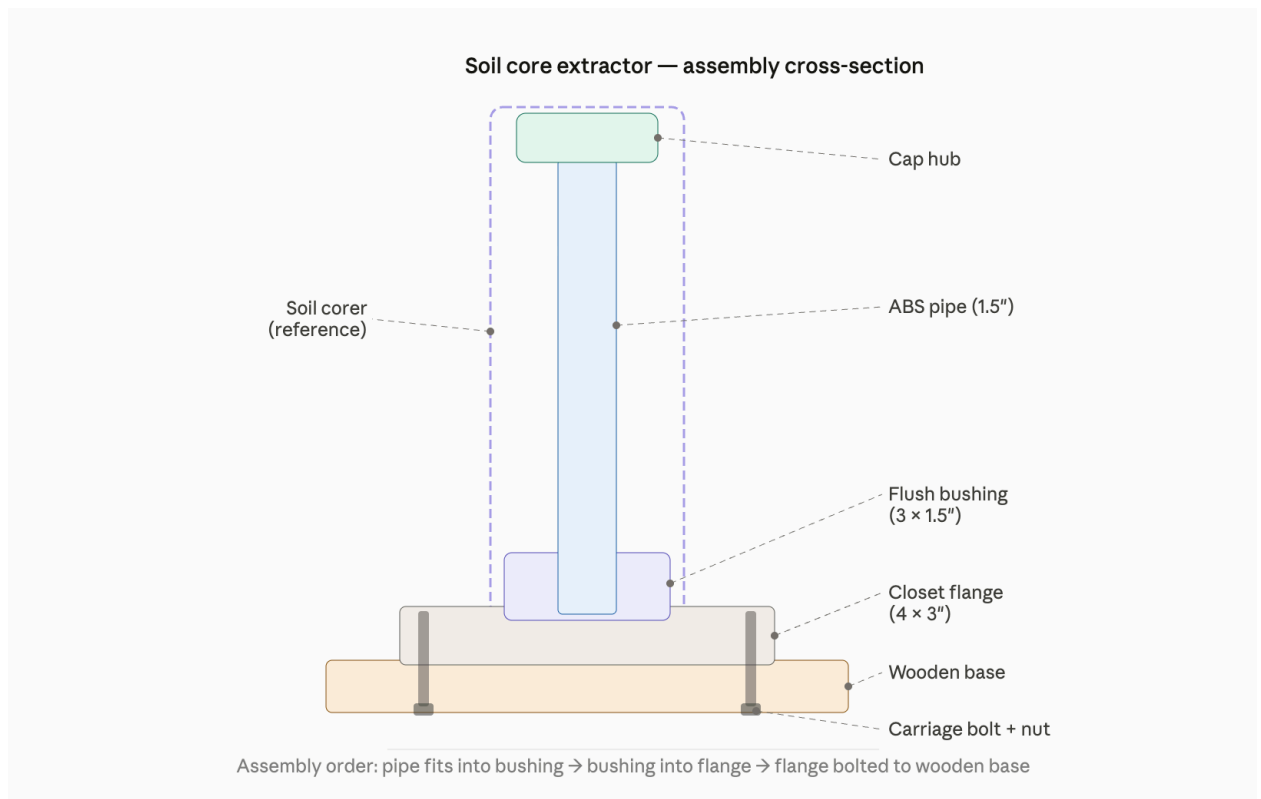
Step 1 - Drill $\frac{1}{4}$ " hole in the center of the cap

Step 2 - Place plug onto top cap, with the large side facing up

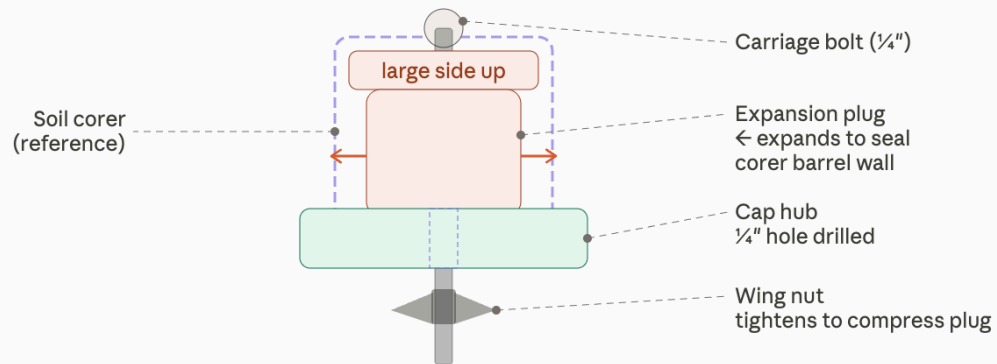
Step 3 - Push the bolt through the top of the plug and through the hole in the cap

Step 4 Secure with a wing nut

Diagrams:



Waterproof plug assembly (optional)



- ① Drill 1/4" hole in centre of cap
- ② Place expansion plug on cap — large side facing up
- ③ Push carriage bolt down through plug and through cap hole
- ④ Thread wing nut from below and tighten — plug expands radially to seal corer